

YANN-EDWIN KETA

Associate professor of physics

 yann-edwin.keta@espci.fr

 yketa.xyz
github.com/yketa

 0000-0001-7736-3676

 Campus P. & M. Curie (Jussieu)
7 quai Saint-Bernard, Paris 5^e

 November 1995

Education

2020-2023

PhD in Physics

Université de Montpellier 

Under the joint supervision of
Ludovic Berthier and Robert L. Jack.

2016-2018

MSc in Physics

École normale supérieure de Lyon 

Specialisation in computational
physics, soft matter, and statistical
physics.

2015-2016

BSc in Physics

École normale supérieure de Lyon 

2013-2015

Classes préparatoires aux grandes écoles (PCSI/PC*)

Lycée Lakanal, Sceaux 

2018-2019 (Gap year)

MA in Social sciences

1st year

École normale supérieure de Lyon 

Skills

Programming

Python , C/C++, GNU/Linux , Git,
symbolic computation (SageMath),
Julia, HTML/JS/CSS, $\LaTeX$.

Molecular dynamics, parallelisation
on CPU and GPU (OpenMP, HOOMD),
biased path ensemble algorithms.

Languages

Français (French) – Native speaker 

English – Fluent 

Nederlands (Dutch) – Beginner 

Interests

- * Extreme music.
- * Free software, open knowledge.
- * Environment protection.
- * French and non-French literature.

Research

Sep 2025 **Associate professor (Maître de conférences)**

- Present UFR de Physique, Faculté des Sciences et Ingénierie,
Sorbonne Université 
Laboratoire de Physique et Mécanique des Milieux Hétérogènes
(PMMH), UMR 7636 CNRS, ESPCI Paris – PSL, Sorbonne Université,
Université Paris Cité 

Oct 2023 **Postdoc: "Physical models of cell sheets"**

- Aug 2025 Instituut-Lorentz for Theoretical Physics,
Universiteit Leiden 

Supervisor: Silke Henkes

Sep 2020 **PhD: "Emergence of disordered collective motion in
dense systems of isotropic self-propelled particles"**

- Sep 2023 Laboratoire Charles Coulomb (L2C), UMR 5221 CNRS,
Université de Montpellier 
Simons collaboration on *Cracking the Glass Problem*

Supervisors: Ludovic Berthier (Montpellier),
Robert L. Jack (Cambridge)

Internships

Oct 2019 **"Large deviations of active particles"**

- July 2020 Department of Applied Mathematics and Theoretical Physics,
University of Cambridge 
Laboratoire Matière et Systèmes Complexes (MSC),
UMR 7057 CNRS, Université Paris Cité 

Supervisors: Robert L. Jack, Michael E. Cates (Cambridge),
Frédéric van Wijland (Paris)

Jan 2018 **"Glassy behaviour in phase-separating active matter"**

- Jul 2018 Stewart Blusson Quantum Matter Institute,
University of British Columbia 

Supervisor: Jörg Rottler

May 2017 **"Jamming criticality of spheroids"**

- Jul 2017 Institutionen för fysik, Umeå universitet 

Supervisor: Peter Olsson

Publications

S. Naik, Y.-E. Keta, K. Pranjic-Ferscha, É. Hannezo, S. Henkes, and C.-P. Heisenberg, "Ker-
atins coordinate tissue spreading by balancing spreading forces with tissue material
properties", *Nature Communications* **17**, 6499 (2026) [DOI:10.1038/s41467-026-72366-
z].  bioRxiv:10.1101/2025.02.14.638262

S. C. Kammeraat, Y.-E. Keta, P. Appleton, I. P. Newton, T. B. Liverpool, R. Sknepnek, I.
Näthke, and S. Henkes, "Correlated cell movements drive epithelial finger formation",
arXiv (2025) [DOI:10.48550/arXiv.2508.01046].  arXiv:2508.01046

Y.-E. Keta and S. Henkes, "Long-range order in two-dimensional systems with fluctu-
ating active stresses", *Soft Matter* **21**, 5710–5719 (2025) [DOI:10.1039/D5SM00208G].
 arXiv:2410.14840

Y.-E. Keta*, J. U. Klamser*, R. L. Jack, and L. Berthier, "Emerging Mesoscale Flows and
Chaotic Advection in Dense Active Matter", *Physical Review Letters* **132**, 218301 (2024)
[DOI:10.1103/PhysRevLett.132.218301].  arXiv:2306.07172

Y.-E. Keta, R. Mandal, P. Sollich, R. L. Jack, and L. Berthier, "Intermittent relaxation and
avalanches in extremely persistent active matter", *Soft Matter* **19**, 3871–3883 (2023)
[DOI:10.1039/D3SM00034F].  arXiv:2212.09836

Y.-E. Keta, R. L. Jack, and L. Berthier, "Disordered collective motion in dense
assemblies of persistent particles", *Physical Review Letters* **129**, 048002 (2022)
[DOI:10.1103/PhysRevLett.129.048002].  arXiv:2201.04902



Y.-E. Keta, É. Fodor, F. van Wijland, M. E. Cates, and R. L. Jack, "Collective motion in large deviations of active particles", *Physical Review E* **103**, 022603 (2021) [DOI:10.1103/PhysRevE.103.022603].  arXiv:2009.07112

Y.-E. Keta and P. Olsson, "Translational and rotational velocities in shear-driven jamming of ellipsoidal particles", *Physical Review E* **102**, 052905 (2020) [DOI:10.1103/PhysRevE.102.052905].  arXiv:2006.05305

T. Marschall, Y.-E. Keta, P. Olsson, and S. Teitel, "Orientational Ordering in Athermally Sheared, Aspherical, Frictionless Particles", *Physical Review Letters* **122**, 188002 (2019) [DOI:10.1103/PhysRevLett.122.188002].  arXiv:1806.01739

Y.-E. Keta and J. Rottler, "Cooperative motion and shear strain correlations in dense 2D systems of self-propelled soft disks", *EPL* **125**, 58004 (2019) [DOI:10.1209/0295-5075/125/58004].

Refereeing

Communications Physics, Nature Communications, Nature Physics, Physical Review E, Physical Review Letters, Scientific Reports, SciPost, Soft Matter.

Responsibilities

- 2025 **Co-organisation of the IntCha26 conference**
- 2026 Institut d'Études Scientifiques de Cargèse 
The conference "Interdisciplinary Challenges in Non-Equilibrium Physics" aims to connect young researchers from diverse backgrounds at the interface between the physics of complex systems and biology.
- 2023 **Co-organisation of the "Smart, Living, and Active Matter" seminar**
- 2025 Universiteit Leiden 
Hosts international speakers between 1 and 4 times a month.

Teaching

- 2025 **"Mathematics", "Physics of continuous media" (Bachelor)**
"Granular materials" (Master)
Sorbonne Université 
Teaching assistant.
- 2025 **"Active Matter" (Advanced Topics Masters Course)**
Universiteit Leiden 
Lecturer. Part of the course "Advanced Topics in Theoretical Physics" from the Dutch Research School of Theoretical Physics.
- 2024 **"Statistical Physics" (Masters)**
Universiteit Leiden 
Teaching assistant.
- 2022 **"Physics for life sciences", "Python for sciences" (Bachelor)**
Université de Montpellier 
Teaching assistant.
- 2018-2019 **Oral interrogator (Physics, Chemistry, Mathematics)**
- 2016-2017 Lycée du Parc, Institution des Chartreux, Lycée La Martinière Diderot (Lyon) 
- 2015 **Volunteer tutor (Physics, Chemistry, Mathematics)**
- 2017 ENSeigner association, École normale supérieure de Lyon 